An AI-Exposure Shock and Entry-Level Preferential Hiring Alternative

Ebba Mark & Claude

Note: Red text indicates the beginning and end of a section generated by Claude.

Note: Blue text indicates either a reminder for Ebba to fix something/double-check Claude's work.

1 Overview

This note describes our candidate alternative for the entry-level policy exercise. Previously, we applied a preferential hiring policy for long-term unemployed and entry-level workers during the simulation window (2001-2024). We will keep the LTU targeting design as it is (run the simulation from 2001-2024 with vs. without the policy), as it provides an interesting commentary on the Work Opportunity Tax Credit (WOTC). Essentially, we ask what could have been the effect of the WOTC if it had been applied effectively? We present this result in line with critiques of the WOTC.

In the original manuscript, we imposed preferential hiring for entry-level workers in the same way (compare 2001-2024 simulation with vs. without the policy). The below alternative takes a forward-looking (and likely more relevant) approach instead. We simulate the model from 2001-2024 without the policy, apply an AI-exposure shock at the end of our simulation period, either with or without the policy. Essentially, rather than testing entry-level preferential hiring against realised business-cycle conditions, it asks what the policy does in response to a hypothetical, occupation-specific demand shock calibrated to recent evidence on AI exposure.

2 Data

2.1 Occupational AI-exposure scores

Occupation-level AI-exposure scores are drawn from **Eloundou, Manning, Mishkin & Rock (2024)**, `https://www.science.org/doi/10.1126/science.adj0998`, which reports LLM-exposure ratings as exposure via direct LLM use plus complementary software) for 923 O*NET-SOC occupations.

Crosswalk. The model's occupation network is coded in Census/ACS 2010 occupation codes (528 occupations). These are crosswalked to 2010 SOC detailed codes via our preferred ACS-SOC crosswalk, then matched against Eloundou et al.'s SOC codes. 361 of 528 occupations (68%) match; of those, 345 (65% of all 528) retain a non-zero exposure score after min-max normalisation to $[0, 1]$ – the remaining 16 matched occupations' raw ratings equal the source's minimum and so normalise to exactly 0. Unmatched occupations are likewise assigned exposure = 0, i.e. treated as unshocked. The result is a single exposure score per occupation, $e_i \in [0, 1]$.

- Ebba to check cross-walk matching to see whether this can improve coverage. For now this crosswalks between 2018 and 2010 codes. I have a crosswalk for this somewhere to make sure it is consistent with other cross-walking we've done in the paper.
- Ebba to check the correlation between exposure shock and our own classification of occupations as entry-level. This will dictate whether we our entry-level hiring policy is effective/relevant here.

3 Method

3.1 Pre-Shock Initialization

The shock is applied at the end of the simulated window (2024). Each model variant considered in this note – the three cross-model variants of Section 3.4 and the full behavioural reference model of Section 4 – is run under its own calibration window, from 2001-01 to 2024-05, for $N = 10$ independent simulations, using that variant's own calibrated γ_u, d_u, θ and behavioural flags. At the end of each run the live network state (every worker's employment status, occupation, wage, and tenure, together with the vacancy pool) is captured. These ten end-states are the starting points for the shock-phase runs below, for that variant.

Because the empirical demand series the calibration is built on ends in 2024, there is no real (read: currently downloaded) demand signal for the post-2024 period. Each occupation's target-demand multiplier is therefore held at its calibrated no-shock value (1.0) going forward, except where reduced by the imposed shock (Section 3.2); the underlying `target_demand` level itself (an absolute headcount anchor used in the model's separation and vacancy-creation rules) is left at its latest simulated value throughout. The real-world vacancy rate is held flat at its last observed value.

3.2 Shock construction

Note: Claude's drafting contribution starts here.

For each occupation i, the target-demand multiplier over the shock horizon $t = 0, \ldots, 60$ months is

$$m_i(t) = 1 - \gamma \cdot e_i \cdot S(t) \tag{1}$$

where e_i is the selected occupation's exposure score (Section 2.1), $\gamma = 0.19$ is the maximum fractional target demand reduction, anchored to the Brynjolfsson et al.'s estimate of the employment gap for young workers in AI-exposed occupations (the paper's ∼19% finding), and $S(t)$ is a logistic ramp normalised to run from exactly 0 at $t = 0$ to exactly 1 at $t = 60$, with midpoint at $t = 30$ (half the horizon) and (uncalibrated!) steepness $8/60$ months^{-1}. Unexposed occupations ($e_i = 0$) are unaffected; the most exposed occupations see target demand fall by up to 19% by the end of the horizon (Brynjolfsson, Chandar, and Chen 2025).

Note: Claude's drafting contribution ends here.

Why γ is needed. e_i measures technical, task-level exposure – e.g. Eloundou et al.'s rating reflects whether LLM assistance could plausibly cut the time needed for a task at equal quality – not a validated demand or employment elasticity. Time savings on a task do not mechanically imply fewer workers are needed (firms may instead expand output), adoption is limited by cost and integration frictions the score does not capture, and most occupations bundle exposed tasks with non-exposed ones (oversight, judgement, interpersonal work) that still require the worker (Eloundou et al. 2024). Applying e_i to target demand with $\gamma = 1$ would assume technical exposure translates one-for-one into demand destruction – an assumption the exposure literature explicitly does not make. $\gamma = 0.19$ instead keeps the shock's ceiling within what has actually been observed in the labour market (the Canaries paper's employment gap for young workers in independently-identified AI-exposed occupations), so e_i sets which occupations are shocked and by how much *relative to one another*, while γ sets the overall magnitude from real-world evidence rather than from the exposure score's own scale.

Figure 1 shows the imposed shock itself: the target-demand multiplier $m_i(t)$, averaged separately across the top quartile of occupations by exposure and the remaining three quartiles.

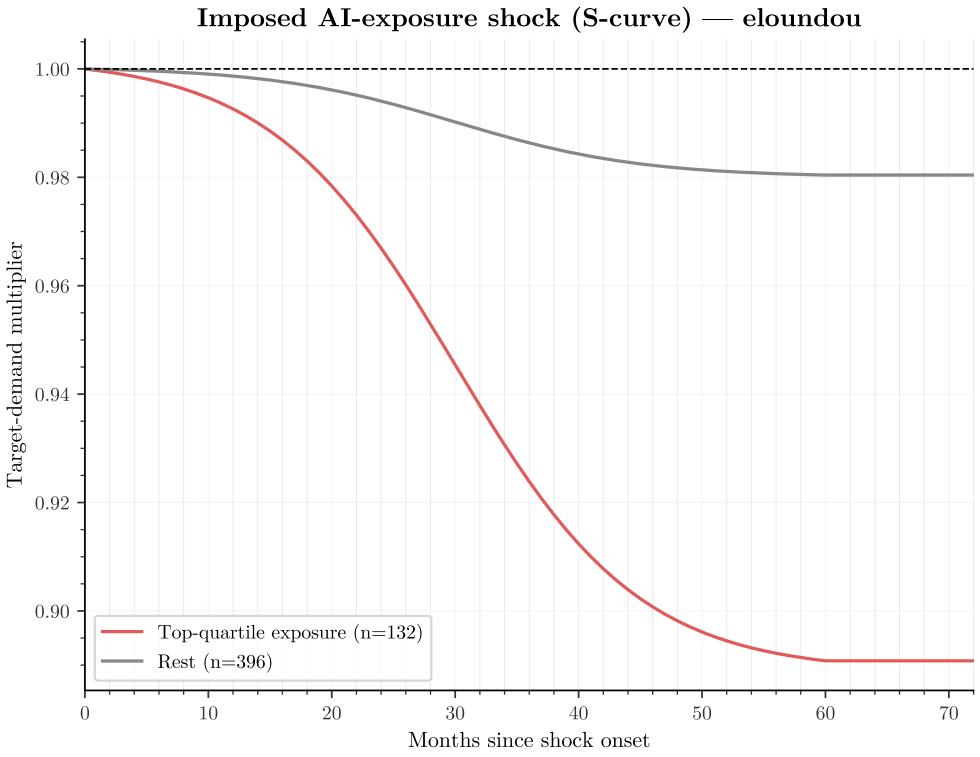


Figure 1: The imposed AI-exposure shock (Eloundou et al. exposure score). Target-demand multiplier $m_i(t)$ over the 60-month shock horizon, averaged across occupations in the top quartile of exposure score versus the remaining three quartiles.

3.3 Shock-phase runs

From each of the ten end-of-simulation states, two 20-year (240-month) simulations are run under the shock:

- **No policy**: `entry_hire_bias` $= 0$.
- **Policy**: `entry_hire_bias` $= 1.0$ – vacancies weight new-entrant applicants twice as heavily as other applicants when selecting a hire.

All other parameters (calibrated γ_u, d_u, θ; behavioural flags; vacancy rule) match the reference model's calibration. Because both scenarios for a given simulation start from the same end-of-simulation states, comparisons below are paired at the simulation level. The S-curve itself still fully ramps in over the first 5 years, exactly as in Section 3.2.

Note: Claude's drafting contribution starts here.

Workers who enter the labour force during that 5-year ramp-in window are tagged, at creation, as a lifetime-earnings cohort (Section 3.5 gives the full mechanics); everyone else is excluded. Fifteen years of follow-up per cohort member was chosen as long enough for a policy-driven earnings gap to emerge and compound well beyond what the near-term figures show, while still keeping the shock-phase runs – repeated for the reference model and all three cross-model variants – within reach computationally. Since the latest-arriving cohort member enters 5 years into the horizon, the total simulated length is set to $20 = 15 + 5$ years so that *every* cohort member, not just the earliest arrivals, gets the same full 15-year follow-up; Figures 2 and 4 both use this full 20-year horizon (tracking 15 years of follow-up per entrant for the latter) – all of it was already simulated and cached for the lifetime-earnings comparison, so showing the full horizon in Figure 2 too costs nothing extra.

Note: Claude's drafting contribution ends here.

3.4 Cross-model comparison

To test whether the policy's effect depends on behavioural assumptions, the same warm-start and shock-phase procedure is repeated independently for the same three model variants we currently use in the paper (non-behavioural benchmark, and two behavioural models without cyclical search with and without reservation wages). Each variant is simulated separately, using its own 2001–2024 history.

Figure 2 shows the policy's effect (policy minus no-policy) over the full 20-year shock horizon for each of the three cross-model variants, for the unemployment rate and failed-hire rate (Panels A-B, entry-level occupations only, matching the primary policy question), and, unlike Panels A-B, computed across *all* occupations so as to pick up spillover effects on the rest of the economy that a purely entry-level metric

would miss: the wage of every currently employed worker (not just workers who were just hired as in Figure 5's wage panel), shown two ways – relative to each occupation's own median wage (Panel C), which isolates within-occupation pay changes from any shift in which occupations people work in, and as a raw dollar level (Panel D), which bundles both together and so is the more direct answer to whether the policy left the workforce better or worse off in pay terms.

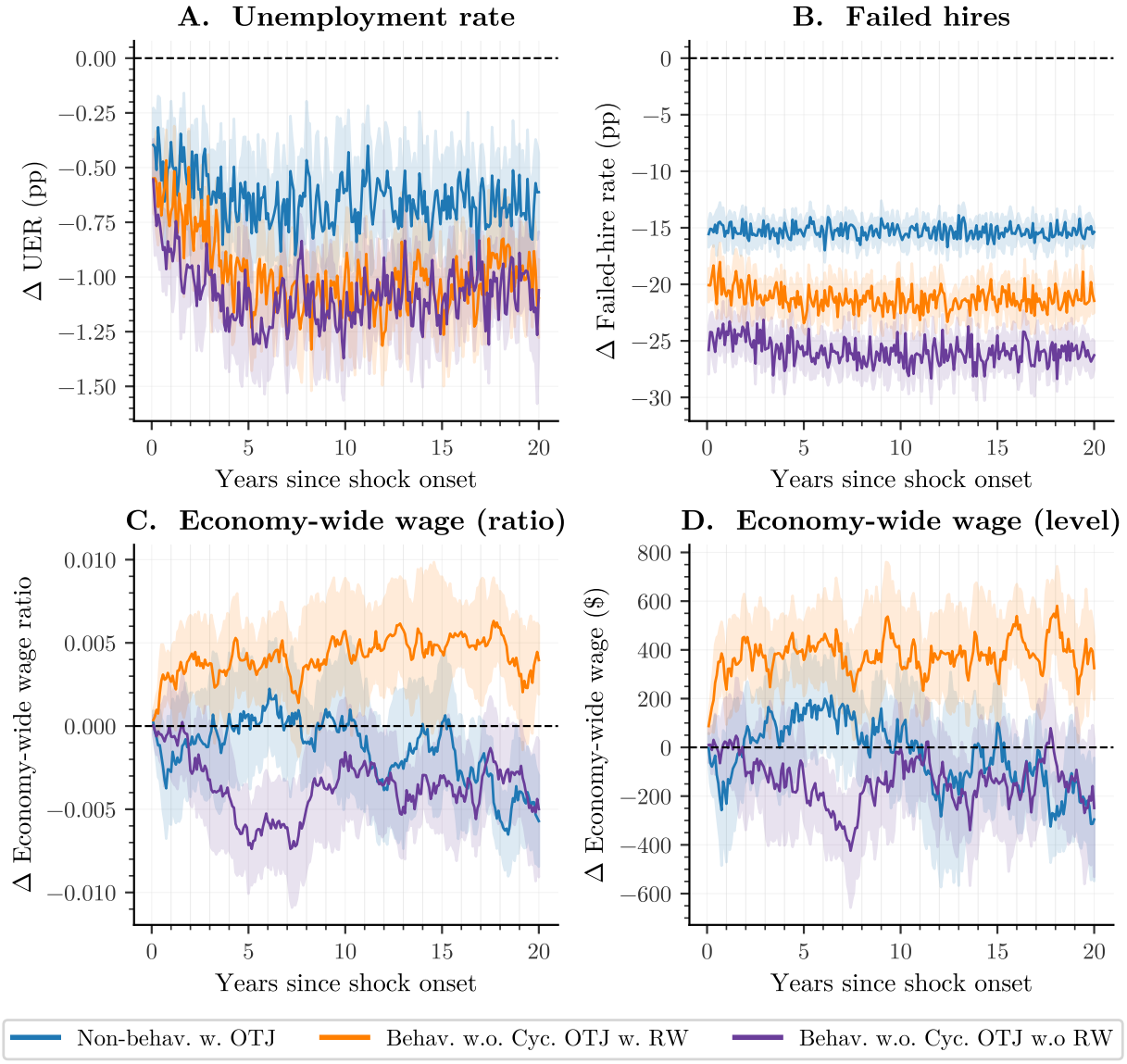


Figure 2: Robustness of the policy effect across behavioural model variants (Eloundou et al. exposure score), over the full 20-year (240-month) shock horizon. Each line shows the policy effect (policy minus no-policy), paired by warm-started simulation and averaged across $N = 10$ simulations per variant. Panels A-B (entry-level occupations only): unemployment rate and failed-hire rate. Panels C-D (*all* occupations, unlike A-B, so as to capture spillover effects on the rest of the economy): the wage of every currently employed worker in scope, not only new hires – Panel C relative to each occupation's own median wage (isolating within-occupation pay changes), Panel D as a raw dollar level (which also reflects any compositional shift in which occupations people work in). Shaded bands show 95% confidence intervals.

3.5 Lifetime earnings of entrants

Note: Claude's drafting contribution starts here.

To ask whether the policy changes entrants' earnings over a longer horizon than the near-term figures capture, we track a cohort of workers who enter the labour force during the shock's 5-year ramp-in window (i.e. while the shock is actually being imposed) and follow their earnings for the 15 years after the window closes. For each such worker, annual snapshots of wage are recorded; a year in which the worker was unemployed at the snapshot date counts as \$0 earned that year rather than being dropped, since excluding it would remove exactly the workers the policy is most likely to affect. Two cumulative measures are tracked per worker: (i) each year's wage divided by the median annual wage of the occupation the worker entered in – fixed once at labour-market entry – summed across years, and (ii) raw annual wage in dollars, summed across years, with no normalisation.

Because both the reference and cross-model shock-phase runs are warm-started from the same end-of-2024 states (Sections 3.1–3.3), the comparison is paired at the level of the warm-started simulation, exactly as in Section 3.4's Δ figure. Concretely, this collapsing happens in two stages, using two different summary statistics. First, within a given simulation and scenario: each cohort worker's cumulative-earnings trajectory (Section 3.5 above) is collapsed to the *median* across every worker in that simulation's cohort, at each year since entry – giving one "typical cohort member" curve per simulation, per scenario. The median (rather than the mean) is used here so that a handful of workers with unusually large cumulative earnings – e.g. an uninterrupted spell in a high-paying occupation – do not dominate the per-simulation summary; with cohorts of hundreds to low thousands of workers per simulation, the median is a stable, well-behaved statistic. Second, because policy and no-policy share the same warm-started starting economy for a given simulation, we take the *difference* between that cohort median under the policy and under no-policy, within that same simulation, at each year; this cancels simulation-specific variation in the warm-started economy rather than letting it compound. Figure 4 plots the *mean* of those ten paired differences (and their 95% CI, from the SEM across the ten simulations), under both measures, for the reference model – the mean (not the median) is used for this second, across-simulation step because $N = 10$ is too small for a median to be a stable or well-characterised estimator. So a given point on the figure is not an individual worker's earnings gain – it is the change in the entrant cohort's *typical (median)* cumulative earnings, averaged across ten independently warm-started economies.

Figure 3 repeats the same procedure for each of the three cross-model variants (Section 3.4), one line per model, styled identically to Figure 2.

Note: this is *not* an individual-level paired comparison. Because `entry_hire_bias` changes how many

random draws `hire()` consumes each period, the two scenarios' shared random-number stream diverges, so which specific occupation each period's entrants land in differs between the policy and no-policy runs even though the total number of entrants per period is identical. The cohort *definition* (arrival window, follow-up length) is the same across both scenarios, and both start from the same warm-started economy, but individual entrants cannot be matched one-to-one across scenarios.

Note: Claude's drafting contribution ends here.

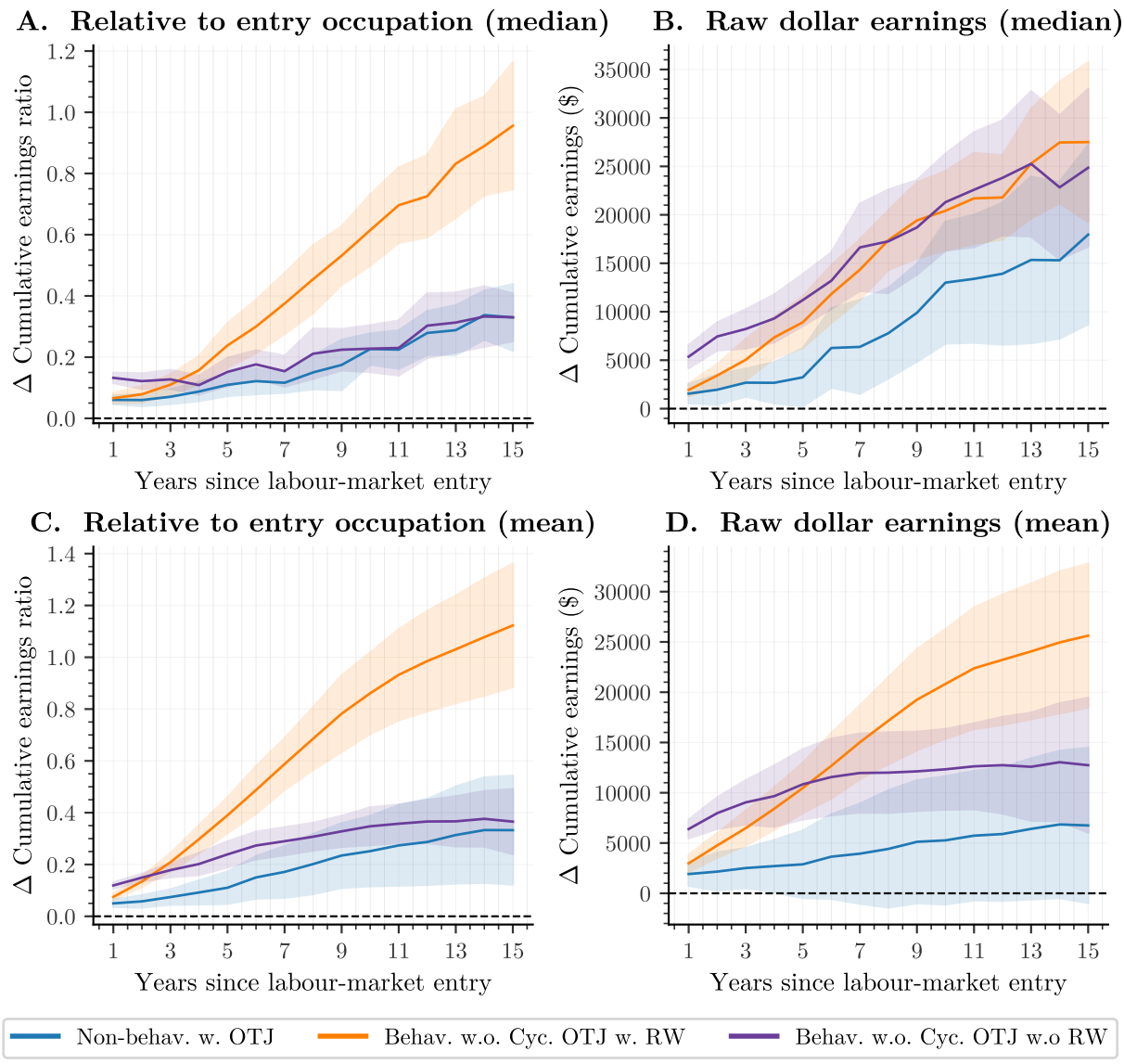


Figure 3: Robustness of the lifetime-earnings effect (Figure 4) across behavioural model variants (Eloundou et al. exposure score). Same cohort, follow-up window, and paired-Δ-by- simulation procedure as Figure 4, repeated independently for each of the three cross-model variants (Section 3.4), one line per model, coloured as in Figure 2. Top row (Panels A-B) uses the per-simulation cohort *median*, the statistic used everywhere else in this note; bottom row (Panels C-D) repeats the same comparison using the per-simulation cohort *mean* instead, shown purely as a robustness check on that choice. Panels A/C: Δ cumulative earnings relative to entry-occupation median wage. Panels B/D: Δ cumulative raw dollar earnings. $N = 10$ simulations per variant, shaded bands show 95% confidence intervals.

4 Single-Model Policy Effect

The reference model used in this section is our "full behavioural model" (cyclical OTJ search intensity, concave search effort, and reservation wages) – the specification used elsewhere in the paper. It is warm-started and shock-tested using exactly the same procedure described in Sections 3.1–3.5, but reported here on its own rather than alongside the three cross-model variants of Section 3.4.

Figure 4 shows the lifetime-earnings comparison described in Section 3.5, for the cohort of entrants who arrive during the shock's ramp-in window, tracked for the 15 years following that window: the policy effect (policy minus no-policy, paired by simulation) under both measures – cumulative earnings relative to entry-occupation median wage (Panel A), and cumulative raw dollar earnings (Panel B).

Figure 5 compares outcomes with (purple) and without (grey) the entry-level hiring policy, for entry-level occupations (solid lines) and the rest of the economy (dotted lines): the unemployment rate, first-job re-employment wages relative to the occupation median, the failed-hire rate (the share of job offers that do not result in a hire), and economy-wide UE and EE transition rates.

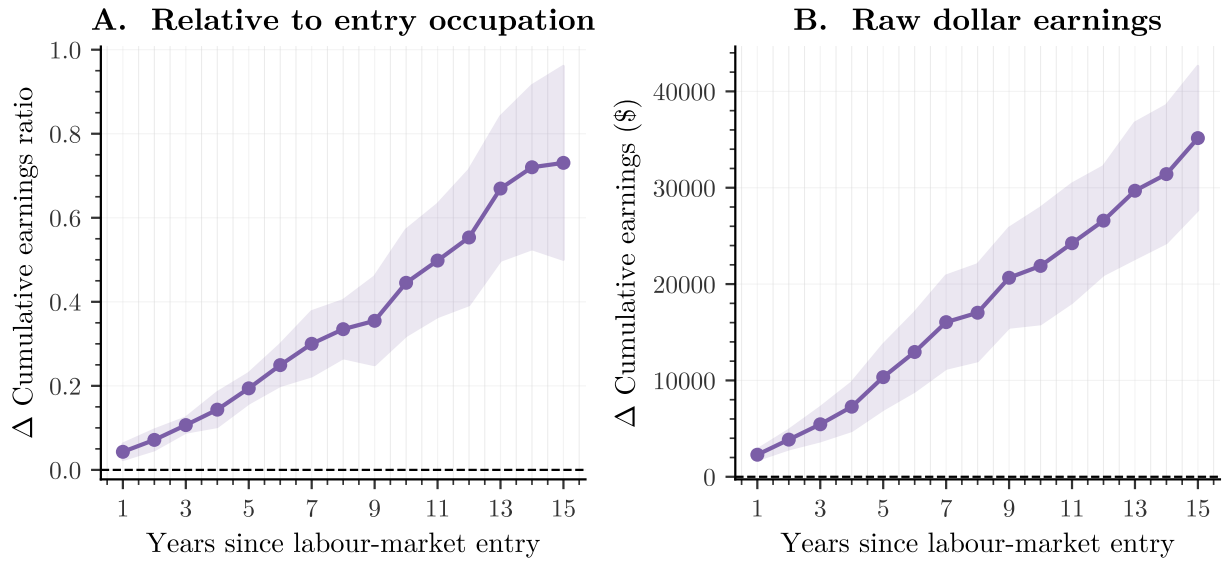


Figure 4: Lifetime-earnings effect of entry-level preferential hiring under the AI-exposure shock (Eloundou et al. exposure score): policy minus no-policy, paired by warm-started simulation. Cohort: workers who enter the labour force during the shock's 5-year ramp-in window, tracked for 15 years following that window. A year unemployed at the annual snapshot counts as \$0 that year. Panel A: Δ cumulative earnings relative to the worker's entry-occupation median wage (fixed at entry, not re-based on later occupation changes – see Section 3.5). Panel B: Δ cumulative raw dollar earnings, no normalisation. Paired at the level of the warm-started simulation, not the individual entrant; $N = 10$ simulations per scenario, shaded bands show 95% confidence intervals.

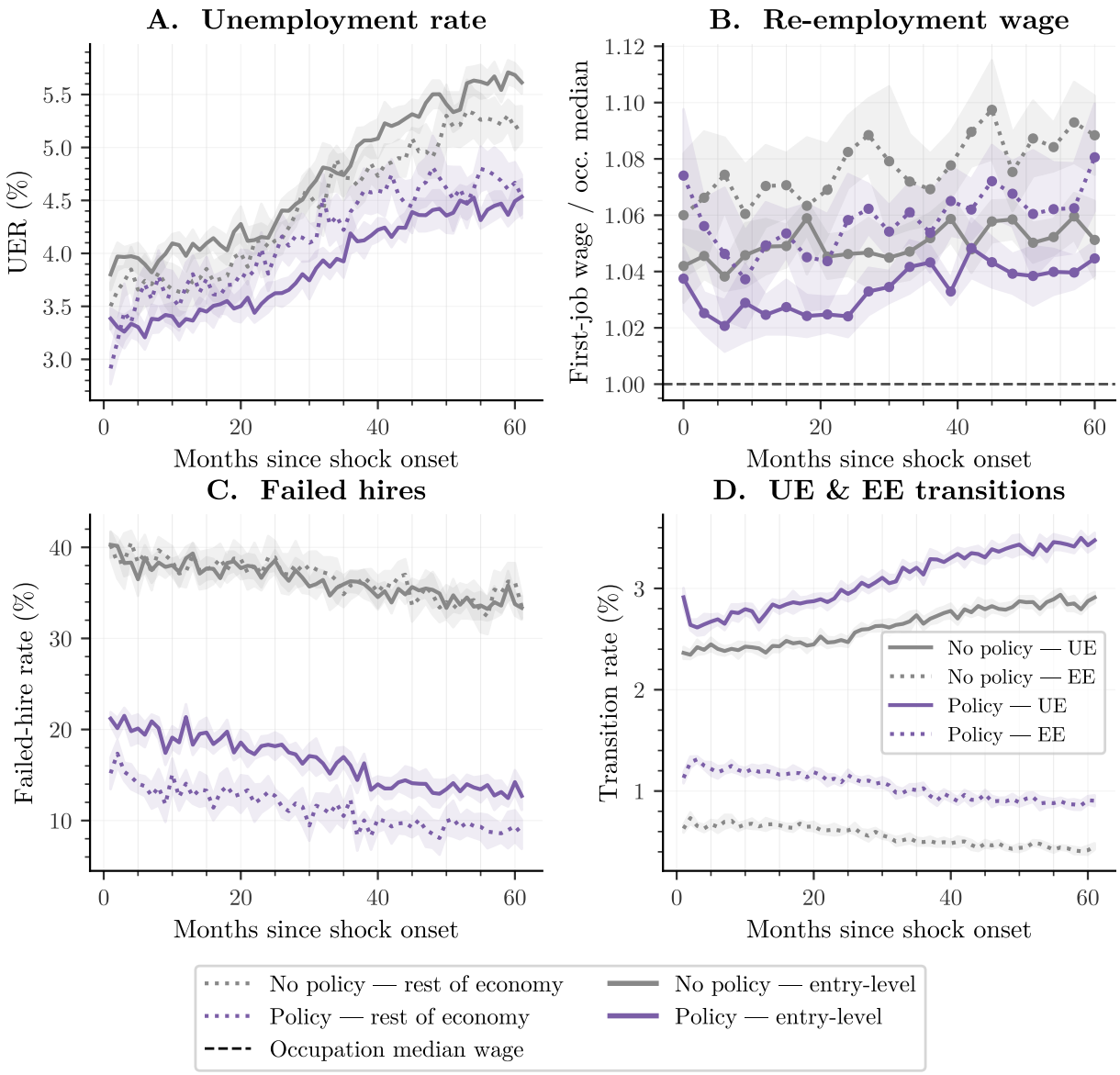


Figure 5: Effect of entry-level preferential hiring under the AI-exposure shock (Eloundou et al. exposure score). Panel A: unemployment rate. Panel B: first-job wage relative to occupation median wage. Panel C: failed-hire rate. Panel D: economy-wide UE and EE transition rates. Solid lines: entry-level occupations; dotted lines (Panels A–C): rest of the economy. Grey: no policy; purple: entry-level preferential hiring. $N = 10$ warm-started simulations per scenario; shaded bands show 95% confidence intervals.

References

Brynjolfsson, Erik, Bharat Chandar, and Ruyu Chen (Nov. 2025). *Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence*. en-US. URL: https://digitaleconomy.stanford.edu/publication/canaries-in-the-coal-mine-six-facts-about-the-recent-employment-effects-of-artificial-intelligence/ (visited on 06/24/2026).

Eloundou, Tyna et al. (June 2024). "GPTs are GPTs: Labor market impact potential of LLMs". In: *Science* 384.6702, pp. 1306–1308. DOI: 10.1126/science.adj0998. URL: https://www.science.org/doi/10.1126/science.adj0998 (visited on 08/28/2026).